EXPERIMENT 8.1.1

Algorithm:-

Step:-1 Start

Step:-2 Input n (dimension of the square matrices).

Step:-3 Print "first matrix:"

Step:-4 Create an empty list A.

Step:-5 Repeat n times:

Read a row of n integers.

Append the row to matrix A.

Step:-6 Print "second matrix:"

Step:-7 Create an empty list B.

Step:-8 Repeat n times:

Read a row of n integers.

Append the row to matrix B.

Step:-9 Create an empty matrix result of size n × n filled with 0.

Step:-10 For i from 0 to n-1

Step:-11 For j from 0 to n-1

Step:-12 For k from 0 to n-1

Step:-13 Compute

result[i][j] = result[i][j] + A[i][k] × B[k][j]

Step:-14 Print "Resultant Matrix:"

Step:-15 Print the result matrix row by row.

Step:-16 Stop

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8.1.1. Multiplication of Two Square Matrices

21:54

Write a Python program to perform the multiplication of two square matrices. The program should take two matrices as input and output their product.

Input Format:

- The first line prints the "dimension: " and reads an integer n, representing the dimensions of the square matrices $(n \times n)$.
- The second line prints: "first matrix:" and then reads n lines, each containing n integers separated by spaces, representing the rows of the first matrix.
- Next line prints: "second matrix:" and then reads n lines, each containing n integers separated by spaces, representing the rows of the second matrix.

Output Format:

- Print "Resultant Matrix:" followed by n lines, each containing n integers separated by spaces, representing the product of the two matrices.

Sample Test Cases +

Explorer

Multiplica...

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```python
#Type Content here....
n =int(input("dimension: "))
print("first matrix:")
A = []
for i in range(n):
    r = list(map(int,input().split()))
    A.append(r)

print("second matrix:")
B=[]
for i in range(n):
```

Debugger

Average time 0.018 s 18.25 ms

Maximum time 0.023 s 23.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 14 ms Debug

Expected output	Actual output
dimension: 2	dimension: 2
first matrix:	first matrix:
1 2	1 2
3 4	3 4
second matrix:	second matrix:
5 6	5 6

Terminal Test cases

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